Laura Caligaris

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Date and place of Birth: 15th of February 1994, Cagliari, Italy
Citizenship: Italian

Biography and Research interests

I am a PhD candidate in Earth, Life and Environmental Sciences at the University of Bologna, with a background in bioinformatics and computational biology. My doctoral research focuses on the characterization of microbial communities in drinking water distribution systems through 16S and 18S rRNA amplicon sequencing, with particular attention to the ecology of opportunistic pathogens such as Legionella spp. and their interactions with eukaryotic hosts. During a visiting period at Eawag (Swiss Federal Institute of Aquatic Science and Technology), I expanded my expertise in cross-kingdom co-occurrence network inference and multivariate community ecology statistics. My broader interests include microbial ecology, metagenomics, and structural bioinformatics.

Publications

2026 **Legionella petroniana sp. nov., a novel species isolated in Bologna, Italy: taxonomic, genomic and ecological insights in the era of environmental change.**
Cristino S, **Caligaris L**, Salaris S, Derelitto C, Bonincontro C, Marino F, Grottola A, Girolamini L.

Improving Microbiological Monitoring of Hospital Surfaces: Loop-Mediated Isothermal Amplification (LAMP) as a New Approach for Rapid Nosocomial Pathogens Detection.
Marino F, Bonincontro C, **Caligaris L**, Bellucci L, Derelitto C, Girolamini L, Cristino S.

2025 **Optimization of Loop-Mediated Isothermal Amplification (LAMP) for the Rapid Detection of Nosocomial Pathogens on Environmental Surfaces**

Marino F, Bonincontro C, **Caligaris L**, Bellucci L, Derelitto C, Girolamini L, Cristino S.

International Journal of Molecular Sciences, 2025, 26, 5933, https://doi.org/10.3390/ijms26135933

Surfaces environmental monitoring of SARS-CoV-2: Loop mediated isothermal amplification (LAMP) and droplet digital PCR(ddPCR) in comparison with standard Reverse-Transcription quantitative polymerase chain reaction (RT-qPCR) techniques

Spiteri S, Salamon I, Girolamini L, Pascale M R, Marino F, Derelitto C, **Caligaris L**, Paghera S, Ferracin M, Cristino S.

PLoS ONE, 2025 20(2): e0317228 http://doi.org/10.131/journal.pone.0317228.

Identification of Legionella by MALDI Biotyper through three preparation methods and an in-house library comparing phylogenetic and hierarchical cluster results

Girolamini L, Caiazza P, Marino F, Pascale M R, **Caligaris L**, Paghera S, Ferracin M, Cristino S.

Scientific Report, 2025 15:2162, https://doi.org/10.1038/s41598-025-85251-4

2024 **An Innovative Approach for the Environmental Monitoring of SARS-CoV-2**

Spiteri S, Marino F, Girolamini L, Pascale M R, Derelitto C, **Caligaris L**, Paghera S, Cristino S.

Pathogens, 2024 13(11), 1022, https://doi.org/10.3390/pathogens13111022

Posters

2026 **The role of a metagenomic approach on water and swab samples during a Pontiac fever epidemiological investigation**

Girolamini L, Bonincontro C, **Caligaris L**, Derelitto C, Sabelli C, Cristino S.

The 36th European Congress of Clinical Microbiology and Infectious Disease (ECCMID) congress, Munich, Germany, 17-21 April 2026

New insights into intraoperative airborne contamination in operating theatres: a multifactorial analysis integrating culture-based identification, metagenomic NGS, particle monitoring, and procedural factors

Lanzoni L, Antonioli P, Campioni D, Girolamini L, **Caligaris L**, Derelitto C, Bonincontro C, Michilli A, Bassi C, Sabbioni S, Kucaj E, Oosthuizen J, Basson H, Brisbane J, Zaffagnini T, Cristino S.

The 36th European Congress of Clinical Microbiology and Infectious Disease (ECCMID) congress, Munich, Germany, 17-21 April 2026

2025 **Microbiome profiling using 16S metagenomics approach to improve water quality assessment**

Caligaris L, Girolamini L, Bonincontro C, Derelitto C, Sabelli C, Cristino S.

ASM Microbe 2025 congress, American Society for Microbiology, United States, Los Angeles (CA), 19-23 June 2025

Environmental Investigation of the First Italian Legionella non-pneumophila outbreak: suggested workflow on sampling and isolates identification

Girolamini L, Spiteri S, Bonincontro C, **Caligaris L**, Derelitto C, La Fauci G, Belletti M, Resi D, Pandolfi P, and Cristino S.

ASM Microbe 2025 congress, American Society for Microbiology, United States, Los Angeles (CA), 19-23 June 2025

Airborne Microbial Investigation in Operating Rooms Using MALDI-TOF and Next-Generation Sequencing (NGS) to Reduce Surgical Site Infection Risk

Lanzoni L, Antonioli P, Bonincontro C, **Caligaris L**, Derelitto C, Girolamini L, Sabbioni S, Cristino S.

ASM Microbe 2025 congress, American Society for Microbiology, United States, Los Angeles (CA), 19-23 June 2025

Evaluation of water quality through pipe microbiome characterization using 16S gene metagenomics approach

Caligaris L, Girolamini L, Marino F, Bonincontro C, Derelitto C, Sabelli C, Cristino S.

The 35th European Congress of Clinical Microbiology and Infectious Disease (ECCMID) congress, Wien, Austria, 11-15 April 2025

2024 **Comparison between cultural method and isothermal loop mediated amplification (LAMP) for hospital environment monitoring**

Marino F, Pascale MR, Girolamini L, Spiteri S, Caiazza P, **Caligaris L**, Derelitto C., Bellucci L, Cristino S.

The 34th European Congress of Clinical Microbiology and Infectious Disease (ECCMID) congress, Barcelona, Spain, 27-30 April 2024

Legionella spp. identification by MALDI-TOF mass spectroscopy: improvement of technique and species relationship

Girolamini L, Caiazza P, Marino F, **Caligaris L**, Pascale M R, Spiteri S, Derelitto C, and Cristino S.

The 34th European Congress of Clinical Microbiology and Infectious Disease (ECCMID) congress, Barcelona, Spain, 27-30 April 2024

Risk factors in endoscopes reprocessing: microbiological surveillance to support the device engineering

Girolamini L, Pascale M R, Marino F, **Caligaris L**, Caiazza P, Spiteri S, Derelitto C, Bisognin F, Dal Monte P and Cristino S.

The 34th European Congress of Clinical Microbiology and Infectious Disease (ECCMID) congress, Barcelona, Spain, 27-30 April 2024

Visiting Periods

October 2025 – April 2026

EAWAG - Swiss Federal Institute of Aquatic Science and Technology – Dübendorf, Switzerland
Project Title: Study of taxonomic, ecological and functional structure of water biofilms in taps through metagenomics data and interaction networks
Supervisor: Dr Hammes, F, Co-supervisor: Dr Gabrielli, M.

Education

November 2023 – Present

University of Bologna – Bologna, Italy
PhD Student in Earth, Life and Environment Sciences
Project Title: NGS and metagenomic applications for in situ evaluation of water supplies contamination according to the water safety plans – EU Directive 2184/2020 and Italian Decree 18/2023, with COPAN Group.
MaB Lab:
https://bigea.unibo.it/it/ricerca/laboratori-di-ricerca/laboratorio-di-microbiologia-ambientale-e-biologia-molecolare-mab

October 2020 – November 2023

University of Rome Tor Vergata – Rome, Italy
Master's degree in Bioinformatics
109/110 Italian grade
Thesis Title: Identification of potential inhibitors of human topoisomerase IB through virtual screening and classical molecular dynamics techniques.
Supervisor: Professor Falconi, M., E-mail: falconi@uniroma2.it

2013 – 2020

University of Pisa – Pisa, Italy
Bachelor's degree in Biological Science
95/100 Italian Grade
Thesis Title: Function and Structure of PET Hydrolase (PETase): an enzyme for plastic's biodegradation.
Supervisor: Professor Camici, M.

Selected coursework

- 27/03/25 - 2/04/2025 - 20/05/2025 *Artificial Intelligence Tools for Content Production in Academia*: Overview of AI tools that can be effectively used in the production of content in the academic field, offered by Prof. Paolo Torroni and Prof. Alessandro Gallo, University of Bologna;
- 08-09/10/24 *Computational Pangenomics*: Pangenome and Pangenomics analysis course, offered by Dr. Andrea Guarracino, Alta Formazione Insubria;
- 11/06/2024 *IR Biotyper*: Basic course on IR Biotyper, offered by Dr. Andrea Baldaccini, Bruker Daltonics;
- 5-7/02/24 *Introduction to Python Programming*: Python language course, offered by Cineca HPC Department;
- 4-6/12/23 *High Performance Bioinformatics*: introduction of cluster machines and bioinformatics optimization pipelines, offered by Cineca HPC Department;

Intership

June 2022 – October 2023

MPV and Human Topoisomerase Virtual Screening
Supervisors: Professor Falconi Mattia (University of Rome Tor Vergata), Dr. Iacovelli Federico (University of Rome Tor Vergata).
In order to identify the best inhibitors of Monkey and Human topoisomerase type IB, Docking simulations and Molecular Dynamics were performed on 160 compounds of interest, known for their drug activity.

November 2022 – January 2023

Sperimental laboratory experience on computational results
Mentor: Professor Fiorani Paola (University of Rome Tor Vergata), Co-Tutor Chiccarella Giulia (University of Rome Tor Vergata).
Analysis of drugs on MPV and Human Topoisomerase: activity assay, religation and relaxation assay.

November 2019 – January 2020

Function and Structure of PET Hydrolase (PETase): an enzyme for plastic's biodegradation
Supervisor: Professor Camici Marcella (University of Pisa).
Overview on Ideonella sakaiensis enzyme: function, structure and modification of key residues to enhance plastic degradation activity.

May 2017 – July 2017

Antibodies analysis in autoimmune diseases
Supervisor: Professor Paolicchi Aldo (University of Pisa).
Execution and analysis of immunological tests on blood samples.

Mentorship and service

16-18 February 2026 **Microbiology Praktikum (Tutor) - ETH Zürich**
Support students on basic microbiology techniques to cultivate and select bacteria species.

November 2022 – June 2023 **Structural Bioinformatic Course (Tutor) - University of Rome Tor Vergata**
Support students during practical lectures involving the use of main bioinformatics softwares and servers.

Other experiences

July 2019 **University of Pisa and Aquilegia association (Botanical Guide)**
Managing and maintenance of Botanical Garden ”Pania di Corfino” and tour guide.

Technical skills

Programming languages
Proficient in: Python, Ruby, C, R.
Basic knowledge of database management systems, MySQL and the SQL language.
Familiar with: HTML5, CSS, PHP.

Softwares

Genomic and Metagenomic Bioinformatics: Assembly, Quality and Annotation tools (FastQC, Trimmomatic, SPAdes, Pilon, Prokka, Busco, Abricate, Snippy, PGGB, Checkm2), Alignment tools (MUMmer, Bwa, Bowtie2, Samtools, Bcftools), Metagenomics and Amplicon Sequencing Analysis (Kraken2, Mothur, Qiime2, DADA2, phyloseq), Taxonomy (Fasttree, Picrust2, iTOL, PR2, GTDB), Co-occurrence network inference (SPIEC-EASI);

Statistical Analysis: Multivariate community ecology statistics (PERMANOVA/adonis2, Procrustes, Mantel test, Spearman correlation), alpha and beta diversity metrics (Shannon, Observed richness, Chao1, Bray-Curtis, Jaccard), data visualization and figure production (ggplot2, patchwork, vegan);

Structural Bioinformatics: Virtual Screening and Drug Design (Autodock Vina), Molecular dynamics and trajectory analysis (AMBER, GROMACS), processing of simulation data using Python programming and Bash scripting;

Graphic visualization software: PyMol, Chimera, VMD, Krona, iTOL;

Scientific databases: Knowledge and use of scientific databases and NCBI (e.g., PubMed, GenBank, Pubchem, PDB, UNIPROT, PR2, GTDB);

Linux OS: Bash scripting for data processing and use of basic Linux OS administration tools and text manipulation commands. Able to work on clusters and optimize computational resources with SLURM Scheduler. Able to work with Conda and Conda environments.

Wet Lab skills

Specific techniques for analyzing the potability of water intended for human consumption according to the directives D.Lgs. 31/2001, DM 14/06/17, EU 2020/2184, D.Lgs. 23/02/2023, and specific ISO standards (UNI EN ISO 19458:2006, UNI EN ISO 6222:2001, ISO 7899-2:2003, UNI EN ISO 16266:2008, UNI EN ISO 9308-1:2017, ISO 16140-2:2016): study of water distribution systems, development of sampling plans, and risk assessment plans in industrial, healthcare, social assistance, community, and dental clinic settings;
Specific techniques for the isolation, culture, and typing of Legionella spp. according to specific regulations: National Guidelines 2015, Regional Guidelines - Emilia-Romagna Region 2018, and specific ISO standard ISO 11731:2017: study of the distribution systems of hot and cold sanitary water for the prevention of Legionella spp., development of sampling plans, and risk assessment plans (DVR) in industrial, healthcare, social assistance, community, and dental clinic settings;
Techniques of spectrophotometric analysis by MALDI-TOF MS using the MALDI Biotyper® System (Bruker); construction of dendrograms; implementation of an in-house database;
Preparation of culture media for the growth of prokaryotic cells;
Extraction of prokaryotic DNA using standard methods or specific kits;
Quantification of nucleic acids (spectrophotometric methods and Qubit/Nanodrop technology);
Storage and disposal of waste according to current regulations.

Languages

English (C1)

- *English*: IELTS course at AngloAmerican Centre of Cagliari,
  via Mameli, n.46

French (A2)

Soft skills

Problem Solving and Organization

Able to quick adapt to project changes and re-organized work.

Communication (Writing/Speaking)

Scientific presentations, exhibitions and workshop organization.

Team Work

Able to manage conflicts and work as part of a team.

Other interests

I am a crochet-aholic girl. I love to create clothes or small handmade presents. In my free time I also enjoy going to the cinema and watching movies, series and animations. I like to walk in nature.